

Galois Theory / MM7043 2026W

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October 08, 2026

1. FIELDS

03.09.2026

Throughout the semester, we will employ the following conventions:

1. Unless otherwise specified, all rings will be assumed to be unital and commutative.
2. A *ring homomorphism* is a map of rings $\varphi : R \rightarrow S$, such that
 - i. $\varphi(x + y) = \varphi(x) + \varphi(y)$, for all $x, y \in R$.
 - ii. $\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y)$, for all $x, y \in R$.
 - iii. $\varphi(1_R) = 1_S$

Definition 1.1. Let R be a ring. An element $x \in R$ is called a *unit* or *invertible*, if there exists an element $y \in R$ such that $x \cdot y = 1$.

The set of all units of R form a group under multiplication and will be denoted R^* .

Example 1.2.

1. $\mathbb{Z}^* = \{-1, 1\}$
2. $\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$
3. For $n \in \mathbb{N}$, it holds that $\mathbb{Z}_n^* = \{m \in \mathbb{Z}_n : \gcd(m, n) = 1\}$.

Definition 1.3. A *field* is a ring K such that $K^* = K \setminus \{0\}$.

Example 1.4. We recall the following fields:

1. $\mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$
2. $\mathbb{Z}_p$, where $p \in \mathbb{P}$.

Lemma 1.5. A ring R is a field if and only if the only ideals of R are $\{0\}$ and R .

Proof. Let $I \subseteq R$ be an ideal. If I contains a unit then $I = R$.

“ $\implies$ ”: Assume R is a field. If $I \subseteq R$ is a non-zero, non-trivial ideal, then I contains a unit, which implies $I = R$.

“ $\impliedby$ ”: If R is not a field, then there exists some non-zero, non-unit element $x \in R$. But then the ideal Rx satisfies $\{0\} \subsetneq Rx \subsetneq R$. ■

Corollary 1.6. *A homomorphism of fields is always injective.*

Proof. Let $\varphi : \mathbb{K} \rightarrow \mathbb{L}$ be a homomorphism of fields. Then $\ker \varphi$ is an ideal in K . We have $1_{\mathbb{K}} \notin \ker \varphi$, therefore $\ker \varphi = \{0\}$ and φ is injective. ■

1.1. Field extensions

When constructing new fields, there are two main approaches:

1. The approach “from above”, where we consider intersections of fields that satisfy our requirements of an extension, which we will discuss in the following.
2. The approach “from below”, where we explicitly construct a new field based on roots of polynomials. We will discuss this approach in the next section.

Lemma 1.7. *Suppose $\mathbb{K}$ is a field and $\{\mathbb{F}_\alpha : \alpha \in I\}$ is a family of subfields of $\mathbb{K}$. Then*

$$\bigcap_{\alpha \in I} \mathbb{F}_\alpha$$

is a subfield of $\mathbb{K}$.

Definition 1.8. Suppose $\mathbb{L}_0 \subseteq \mathbb{L}$ are fields and $\alpha_1, \alpha_2, \dots \in \mathbb{L}$. Then we define

$$\mathbb{L}_0(\alpha_1, \alpha_2, \dots)$$

as the intersection of all subfields of $\mathbb{L}$ that contain $\mathbb{L}_0$ and $\alpha_1, \alpha_2, \dots$, and will be called the *extension* of $\mathbb{L}_0$ generated by $\alpha_1, \alpha_2, \dots$.

It is clear that such an extension is the smallest subfield of $\mathbb{L}$ that contains $\mathbb{L}_0$ and the elements $\alpha_1, \alpha_2, \dots$.

Example 1.9.

1. Consider the fields $\mathbb{Q} \subseteq \mathbb{R}$, then one can show

$$\mathbb{Q}(\sqrt{2}) = \{p + q\sqrt{2} : p, q \in \mathbb{Q}\}.$$

2. Since the element $1_{\mathbb{Q}}$ generates $\mathbb{Q}$ as a field, it follows that $\mathbb{Q}$ has no proper subfields, i.e. $\mathbb{Q}$ is a *primary field*.
3. The field $\mathbb{F}_p$ is a primary field for any $p \in \mathbb{P}$.

Definition 1.10. Let $\mathbb{F}$ be a field. The *primary subfield* of $\mathbb{F}$ is the subfield generated by the element $1_{\mathbb{F}}$.

Definition 1.11. Let R be a ring, then its *characteristic* is defined by

$$\text{char } R := \begin{cases} \min\left\{n \in \mathbb{N} : \sum_{j=1}^n 1_R = 0\right\}, & \text{if such an } n \text{ exists,} \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 1.12. Let $\mathbb{F}$ be a field.

1. If $\text{char } \mathbb{F} = 0$, then the primary subfield of $\mathbb{F}$ is^[1] $\mathbb{Q}$.
2. If $\text{char } \mathbb{F} = p \in \mathbb{P}$, then the primary subfield of $\mathbb{F}$ is $\mathbb{F}_p$.

Lemma 1.13. Suppose $M \subsetneq R$ is a maximal ideal. Then the quotient R/M is a field.

Remark 1.14. Suppose $\mathbb{K}$ is a field and consider $\mathbb{K}[x]$, the ring of polynomials in x with coefficients in $\mathbb{K}$. Recall that $\mathbb{K}[x]$ is a principle ideal domain. A non-trivial ideal $(f(x))$ is maximal in $\mathbb{K}[x]$ if and only if $f(x)$ is an irreducible polynomial. In particular, $\mathbb{K}[x]/(f(x))$ is a field whenever $f(x)$ is irreducible over $\mathbb{K}$.

Moreover, $\mathbb{K}[x]/(f(x))$ contains $\mathbb{K}$ as a subfield, via the homomorphisms

$$\mathbb{K} \hookrightarrow \mathbb{K}[x] \rightarrow \mathbb{K}[x]/(f(x)).$$

Therefore, for every irreducible $f(x)$ the field $\mathbb{K}[x]/(f(x))$ is a field extension of $\mathbb{K}$.

Note that $f(x)$ being irreducible with $\deg f > 1$ implies that $f(x)$ has no roots in $\mathbb{K}$. But $f(x)$ always has a root in $\mathbb{K}[x]/(f(x))$, because $f(x)$ is zero as an element of $\mathbb{K}[x]/(f(x))$.

The above demonstrates, roughly, that there are relationships:

$$\begin{array}{ccc} \text{Polynomials} & \longleftrightarrow & \text{Field extensions} \\ \text{Properties of roots} & \longleftrightarrow & \text{Symmetries of field extensions} \end{array}$$

Practically, we want some criteria to determine if a polynomial is irreducible.

^[1]We always mean “up to isomorphism”.

Remark 1.15. If $\mathbb{K}$ is a field and $p(x) \in \mathbb{K}[x]$, $\deg p > 1$ is a polynomial with a root in $\mathbb{K}$, then $p(x)$ is not irreducible. Indeed, for any $a \in \mathbb{K}$ one can write

$$p(x) = q(x)(x - a) + p(a)$$

for some polynomial $q(x) \in \mathbb{K}[x]$. If a is a root of $p(x)$, this demonstrates that $p(x)$ is not irreducible.

Note that the converse fails: The polynomial $(x^2 + 1)^2$ has no roots in $\mathbb{R}$, yet is not irreducible.

Lemma 1.16. *For a field $\mathbb{K}$, a polynomial $p(x) \in \mathbb{K}[x]$ with $2 \leq \deg p \leq 3$ is irreducible if and only if it has no roots in $\mathbb{K}$.*

Lemma 1.17 (Gauss). *Suppose $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{Z}[x]$ is primitive, that is $\gcd(a_0, a_1, \dots, a_n) = 1$. Then $p(x)$ is irreducible in $\mathbb{Z}[x]$ if and only if it is irreducible in $\mathbb{Q}[x]$.*

Lemma 1.18 (Eisenstein's Criterion). *Let $q(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{Z}[x]$ and suppose there exists a prime $p \in \mathbb{P}$ such that:*

1. $p \nmid a_n$
2. $p \mid a_0, \dots, a_{n-1}$
3. $p^2 \nmid a_0$

Then $q(x)$ is irreducible in $\mathbb{Q}[x]$.

Example 1.19.

1. The polynomial

$$q(x) = 2x^3 + 6x^2 - 9x + 15$$

satisfies Eisenstein's criterion with $p = 3$.

2. The polynomial

$$q(x) = x^2 + x + 1$$

does not satisfy Eisenstein's criterion. Nevertheless, one can consider the transformation

$$r(x) = q(x + 1) = (x + 1)^2 + (x + 1) + 1 = x^2 + 3x + 3,$$

which does satisfy the criterion. Hence $q(x)$ is also irreducible.

More generally, using the same method as above, one can show that for any prime $p \in \mathbb{P}$, the polynomial

$$q(x) = x^{p-1} + x^{p-2} + \dots + 1$$

is irreducible.

Example 1.20. We recall the following basic homomorphisms:

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1. Suppose R is a ring. Then there exists a unique ring homomorphism

$$\varepsilon : \mathbb{Z} \rightarrow R, \quad n \mapsto n \cdot 1_R.$$

2. Suppose $\mathbb{F}$ is a field. Let $\varepsilon : \mathbb{Z} \rightarrow \mathbb{F}$ be the ring homomorphism from the above, and let $I = \ker \varepsilon$. Then ε induces a monomorphism $\mathbb{Z}/I \rightarrow \mathbb{F}$. In particular, $\mathbb{Z}/I$ is an integral domain, therefore $I \subseteq \mathbb{Z}$ is a prime ideal. Thus $I = \{0\}$ or $I = (p)$, for some $p \in \mathbb{P}$. This “formalizes” our previous definition of $\text{char } \mathbb{F}$.

- i. If $\text{char } \mathbb{F} = 0$, then $\varepsilon : \mathbb{Z} \rightarrow \mathbb{F}$ is a monomorphism. In particular, ε extends to a monomorphism $\mathbb{Q} \rightarrow \mathbb{F}$.

- ii. If $\text{char } \mathbb{F} = p \in \mathbb{P}$, then $\text{ran } \varepsilon \cong \mathbb{Z}/p\mathbb{Z} \cong \mathbb{F}_p$.

3. Suppose we have a fixed homomorphism of rings $\iota : R \rightarrow S$. Then for every choice of $\theta \in S$ there exists a unique ring homomorphism

$$\text{ev}_\theta : R[x] \rightarrow S$$

such that $\text{ev}_\theta(r) = \iota(r)$, for all $r \in R$, and $\text{ev}_\theta(x) = \theta$, namely the evaluation map $p(x) \mapsto p(\theta)$.

Definition 1.21. Let $\mathbb{F}$ be a field. A *(field) extension* of $\mathbb{F}$ is a field $\mathbb{K}$ containing^[2] $\mathbb{F}$ as a subfield. We will also write $\mathbb{F} \hookrightarrow \mathbb{K}$ to denote this relationship.

Example 1.22.

1. $\mathbb{Q} \hookrightarrow \mathbb{R}$
2. $\mathbb{R} \hookrightarrow \mathbb{C}$

If $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension, then $\mathbb{K}$ can be viewed as a vector space over $\mathbb{F}$.

Definition 1.23. The *degree* of a field extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is defined as the dimension of $\mathbb{K}$ as a vector space over $\mathbb{F}$. We also denote this as $[\mathbb{K} : \mathbb{F}]$ or $\dim_{\mathbb{F}} \mathbb{K}$.

Example 1.24.

^[2]Again, up to a monomorphism.

1. $[\mathbb{R} : \mathbb{Q}] = \infty$
2. $[\mathbb{C} : \mathbb{R}] = 2$

Lemma 1.25. Suppose we have field extensions $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$, then

$$[\mathbb{L} : \mathbb{F}] = [\mathbb{L} : \mathbb{K}] \cdot [\mathbb{K} : \mathbb{F}].$$

Definition 1.26. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension and $\alpha \in \mathbb{K}$. We say that α is *algebraic* over $\mathbb{F}$ if there exists a non-zero polynomial $p(x) \in \mathbb{F}[x]$ such that $p(\alpha) = 0$. Otherwise α is called *transcendental* over $\mathbb{F}$.

Example 1.27.

1. $\sqrt{2}, \sqrt{3}$ are algebraic over $\mathbb{Q}$.
2. π, e are transcendental over $\mathbb{Q}$.

Definition 1.28. We say that an extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is *algebraic*, if every element $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$. Otherwise, we say that the extension is *transcendental*.

1.2. Minimal polynomials

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension and $\alpha \in \mathbb{K}$ is an algebraic element over $\mathbb{F}$. Then

$$I = \{p(x) \in \mathbb{F}[x] : p(\alpha) = 0\}$$

is a nonzero ideal in $\mathbb{F}[x]$. Since $\mathbb{F}[x]$ is a principle ideal domain, there exists a unique monic polynomial $m_{\alpha, \mathbb{F}}(x) \in \mathbb{F}[x]$, such that $I = (m_{\alpha, \mathbb{F}}(x))$.

Definition 1.29. The polynomial $m_{\alpha, \mathbb{F}}(x)$ in the above is called the *minimal polynomial* of α over $\mathbb{F}$. If $\mathbb{F}$ is clear from the context, we also abbreviate this as $m_{\alpha}(x)$.

Lemma 1.30. A monic polynomial $p(x) \in \mathbb{F}[x]$ is a minimal polynomial of $\alpha \in \mathbb{K}$ if and only if

1. $p(\alpha) = 0$, and
2. $p(x)$ is irreducible over $\mathbb{F}$.

Proof.

“ $\implies$ ”: Suppose $p(x) = m_{\alpha}(x)$. By definition, $p(\alpha) = 0$ and $p(x)$ generates the ideal of polynomials vanishing at α . Suppose $p(x) = q_1(x) \cdot q_2(x)$. Then $p(\alpha) = q_1(\alpha) \cdot q_2(\alpha)$.

Without loss of generality, assume that $q_1(\alpha) = 0$, that is $q_1(x) \in (p(x))$, therefore

$p(x) \mid q_1(x)$. But also $q_1(x) \mid p(x)$ by our assumption. Therefore $p(x)$ and $q_1(x)$ are associate in $\mathbb{F}[x]$, hence $q_2(x)$ is a unit.

“ $\Leftarrow$ ”: Similarly to the above. ■

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$. Since $m_{\alpha, \mathbb{F}}(x)$ is irreducible, the ideal $(m_{\alpha, \mathbb{F}}(x))$ is maximal. Hence $\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x))$ is a field.

Proposition 1.31. *In the context above, $\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x)) \cong \mathbb{F}(\alpha)$.*

Proof. Consider the evaluation homomorphism

$$\text{ev}_\alpha : \mathbb{F}[x] \rightarrow \mathbb{K}.$$

The image of ev_α is $\mathbb{F}[\alpha]$, the subring of $\mathbb{K}$ generated by $\mathbb{F}$ and α . But the image is also isomorphic to $\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x)) = \mathbb{F}[x]/\ker \text{ev}_\alpha$, which is a field. ■

Corollary 1.32. *In the context above, $[\mathbb{F}(\alpha) : \mathbb{F}] = \deg m_{\alpha, \mathbb{F}}$.*

Proof. This follows from the general fact that, for a monic polynomial $p(x) = a_0 + \dots + a_{n-1}x^{n-1} + x^n \in \mathbb{F}[x]$, it holds that

$$\mathbb{F}[x]/(f(x)) \cong \{b_0 + b_1x + \dots + b_{n-1}x^{n-1} : b_0, \dots, b_{n-1} \in \mathbb{F}\} \cong \mathbb{F}^n. \quad \blacksquare$$

Note that conversely, if $p(x) = a_0 + \dots + a_{n-1}x^{n-1} + x^n \in \mathbb{F}[x]$ is an irreducible polynomial, then $\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x))$ is a field extension of $\mathbb{F}$ of degree n . Furthermore, the element $x + (p(x)) \in \mathbb{F}[x]/(p(x))$ is a zero of $p(x)$, and $p(x)$ is the minimal polynomial of the element $x + (p(x))$.

Lemma 1.33. *Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension, then $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$ if and only if $[\mathbb{F}(\alpha) : \mathbb{F}] < \infty$.*

Proof.

“ $\Rightarrow$ ”: Follows immediately from the previous corollary.

“ $\Leftarrow$ ”: The elements $1, \alpha, \alpha^2, \dots, \alpha^n$ are linearly dependant over $\mathbb{F}$. Therefore there exist $s_0, s_1, \dots, s_n \in \mathbb{F}$, not all zero, such that $s_0 + s_1\alpha + \dots + s_n\alpha^n = 0$, therefore α is algebraic over $\mathbb{F}$. ■

Corollary 1.34. *Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension. If $[\mathbb{K} : \mathbb{F}] < \infty$, then $\mathbb{K}$ is algebraic over $\mathbb{F}$.*

Corollary 1.35. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $\alpha, \beta \in \mathbb{K}$ are algebraic over $\mathbb{F}$. Then $\alpha \pm \beta, \alpha\beta, \frac{1}{\alpha}$ (provided that $\alpha \neq 0$) are algebraic over $\mathbb{F}$.*

Proof. By the previous lemma, it suffices to show that the respective generated extensions are of finite degree. The fields $\mathbb{F}(\alpha \pm \beta), \mathbb{F}(\alpha\beta)$ are subfields of $\mathbb{F}(\alpha, \beta)$. But the latter extension has finite degree, since

$$[\mathbb{F}(\alpha, \beta) : \mathbb{F}] = [\mathbb{F}(\alpha, \beta) : \mathbb{F}(\alpha)] \cdot [\mathbb{F}(\alpha) : \mathbb{F}] \leq [\mathbb{F}(\beta) : \mathbb{F}] \cdot [\mathbb{F}(\alpha) : \mathbb{F}] < \infty.$$

The final claim follows from the fact that $\mathbb{F}(\alpha) = \mathbb{F}(1/\alpha)$. ■

Corollary 1.36. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$. Then the set of all algebraic elements of $\mathbb{F}$ is a sub-extension of $\mathbb{F}$, in other words it is a sub-field of $\mathbb{K}$ that is also an extension of $\mathbb{F}$. This is called the algebraic closure of $\mathbb{F}$ in $\mathbb{K}$.*

Example 1.37. We want to determine the minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$. We consider that we ultimately want something like $x = \sqrt{2} + \sqrt{3}$. Squaring yields $x^2 + 5 + 2\sqrt{6}$, or equivalently $x^2 - 5 = 2\sqrt{6}$. Squaring again yields $x^4 - 10x^2 + 1$. It remains to show that this polynomial, say $p(x)$, is minimal, which will be the case if we can show that it is irreducible over $\mathbb{Q}$.

Since $p(x)$ is monic with integer coefficients, by Gauss' lemma it suffices to rule out factorizations over $\mathbb{Z}$. Note that, by the rational root theorem, the only root candidates are ± 1 , and neither turn out to actually be roots. Hence there can be no linear factors.

Suppose that

$$\begin{aligned} p(x) &= x^4 - 10x^2 + 1 \\ &\stackrel{?}{=} (x^2 + ax + b)(x^2 + cx + d) \\ &= x^4 + ax^3 + bx^2 + cx^3 + acx^2 + bcx + dx^2 + adx + bd \\ &= x^4 + (a + c)x^3 + (b + ac + d)x^2 + (ad + bc)x + bd \end{aligned}$$

with $a, b, c, d \in \mathbb{Z}$. By comparing coefficients, we get

- (i) $a + c = 0$,
- (ii) $b + ac + d = -10$,
- (iii) $ad + bc = 0$,
- (iv) $bd = 1$.

From (i) we get $a = -c$, from (iii) we get $a(d - b) = 0$.

- If $a = 0$, then $c = 0$ and from (ii) we get $d = -10 - b$. From (iv) we get

$$1 \stackrel{?}{=} b(-10 - b) = -10b - b^2 \iff b^2 + 10b - 1 = 0,$$

which yields $b = -5 \pm \sqrt{96} = -5 \pm 2\sqrt{6} \notin \mathbb{Q}$.

- If $d = b$, then from (iv) we get $1 = b^2$, thus $b = \pm 1$, and from (ii) we get $a^2 - 2b = 10$. The case $b = 1$ yields $a^2 = 12$, while $b = -1$ yields $a^2 = 8$, both of which are not perfect squares.

Therefore we have shown that $p(x)$ is irreducible over $\mathbb{Q}$, and hence the minimal polynomial of $\sqrt{2} + \sqrt{3}$.

Remark 1.38. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension and $\alpha_1, \dots, \alpha_m \in \mathbb{K}$ are algebraic. Then $\mathbb{F}(\alpha_1, \dots, \alpha_m)$ is a finite degree (and therefore algebraic) extension of $\mathbb{F}$.

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Lemma 1.39. If $\mathbb{F} \hookrightarrow \mathbb{K}$ and $\mathbb{K} \hookrightarrow \mathbb{L}$ are algebraic extensions, then the extension $\mathbb{F} \hookrightarrow \mathbb{L}$ is also algebraic.

Proof. Let $\alpha \in \mathbb{L}$, then α is algebraic over $\mathbb{K}$. Therefore there exists a polynomial $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{K}[x]$ such that $p(\alpha) = 0$. Note that all $a_0, \dots, a_n$ are algebraic over $\mathbb{F}$. Furthermore, $p(x) \in \mathbb{F}(a_0, \dots, a_n)[x]$, so α is algebraic over $\mathbb{F}(a_0, \dots, a_n)$. Therefore

$$[\mathbb{F}(a_0, \dots, a_n, \alpha) : \mathbb{F}] = [\mathbb{F}(a_0, \dots, a_n, \alpha) : \mathbb{F}(a_0, \dots, a_n)] \cdot [\mathbb{F}(a_0, \dots, a_n) : \mathbb{F}] < \infty. \quad \blacksquare$$

1.3. Splitting fields

Definition 1.40. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension and $p(x) \in \mathbb{F}[x]$, $\deg p \geq 1$. We say that $p(x)$ splits completely over $\mathbb{K}$ if $p(x)$ is a product of linear polynomials over $\mathbb{K}$.

Example 1.41.

1. The polynomial $x^2 - 2 \in \mathbb{Q}[x]$ splits completely over $\mathbb{R}$, since

$$x^2 - 2 = (x + \sqrt{2})(x - \sqrt{2}).$$

2. The polynomial $x^2 + 2 \in \mathbb{Q}[x]$ splits completely over $\mathbb{C}$, since

$$x^2 + 2 = (x + i\sqrt{2})(x - i\sqrt{2}).$$

Remark 1.42. If $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$ and $p(x) \in \mathbb{F}[x]$ splits completely over $\mathbb{K}$, then it also splits completely over $\mathbb{L}$.

Remark 1.43. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{F}[x]$. Then $p(x)$ splits completely over $\mathbb{K}$ if and only if there exist $\alpha_1, \dots, \alpha_n \in \mathbb{K}$ such that

$$p(x) = a_n(x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n).$$

Note that $\{\alpha_1, \dots, \alpha_n\}$ is uniquely determined by $p(x)$.

Suppose $p(x) \in \mathbb{F}[x]$ splits completely over $\mathbb{K}$ as above. Consider the sub-extension $\mathbb{F} \hookrightarrow \mathbb{F}(\alpha_1, \dots, \alpha_n) \hookrightarrow \mathbb{K}$. This intermediate extension has the following properties:

1. $p(x)$ splits completely over $\mathbb{F}(\alpha_1, \dots, \alpha_n)$.
2. For any proper sub-extension $\mathbb{F} \hookrightarrow \mathbb{L} \hookrightarrow \mathbb{F}(\alpha_1, \dots, \alpha_n)$, $p(x)$ does not split completely over $\mathbb{L}$.

Definition 1.44. Let $p(x) \in \mathbb{F}[x] \setminus \mathbb{F}$. A *splitting field* over $\mathbb{F}$ is any field extension $\mathbb{F} \hookrightarrow \mathbb{K}$ such that:

1. $p(x)$ splits completely over $\mathbb{K}$.
2. $p(x)$ does not split completely over any proper sub-extension $\mathbb{F} \hookrightarrow \mathbb{L} \hookrightarrow \mathbb{K}$.

We have already shown:

Lemma 1.45. If $\mathbb{F} \hookrightarrow \mathbb{L}$ is an extension such that $p(x) \in \mathbb{F}[x] \setminus \mathbb{F}$ splits completely over $\mathbb{L}$, then $\mathbb{L}$ contains a splitting field of $p(x)$.

Proposition 1.46. For any $p(x) \in \mathbb{F}[x]$, $\deg p \geq 1$, there exists an extension $\mathbb{F} \hookrightarrow \mathbb{K}$ such that $p(x)$ splits completely over $\mathbb{K}$. It follows that every polynomial has a splitting field.

Proof. Suppose $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{F}[x]$. We argue by induction on n .

- If $n = 1$, then $p(x)$ splits completely over $\mathbb{F}$.
- Suppose $n > 1$ and assume that any polynomial of degree $n - 1$ over any field splits completely over some field extension. If $p(x)$ splits completely over $\mathbb{F}$ there is nothing to show. So assume that $p(x)$ does not split completely over $\mathbb{F}$. Then we can write (in $\mathbb{F}[x]$) that $p(x) = q(x) \cdot r(x)$, where $q(x)$ is irreducible over $\mathbb{F}$ and $\deg q > 1$. View $q(x)$ as a minimal polynomial of a root α and consider the induced extension $\mathbb{F} \hookrightarrow \mathbb{L}$. By assumption, we can write $q(x) = (x - \alpha) \cdot q_1(x)$. Consider the polynomial $q_1(x) \cdot$

$r(x) \in \mathbb{L}[x]$, then $\deg(q_1 \cdot r) = n - 1$. By induction hypothesis, there is an extension $\mathbb{L} \hookrightarrow \mathbb{K}$ such that $q_1(x) \cdot r(x)$ splits completely over $\mathbb{K}$. So we have extensions $\mathbb{F} \hookrightarrow \mathbb{L} \hookrightarrow \mathbb{K}$ and $p(x) = (x - \alpha) \cdot q_1(x) \cdot r(x)$. It follows that $p(x)$ splits completely over $\mathbb{K}$. ■

Remark 1.47. Suppose $\varphi : \mathbb{F} \rightarrow \mathbb{K}$ is a field isomorphism. Then φ induces a ring isomorphism

$$\varphi_x : \mathbb{F}[x] \rightarrow \mathbb{K}[x], p(x) \mapsto \varphi(p(x)).$$

A polynomial $p(x) \in \mathbb{F}[x]$ is irreducible if and only if $\varphi_x(p(x))$ is irreducible in $\mathbb{K}[x]$.

For any polynomial $p(x) \in \mathbb{F}[x]$, φ also induces an isomorphism of quotient rings

$$\mathbb{F}[x]/(p(x)) \cong \mathbb{K}[x]/(\varphi(p(x))).$$

Theorem 1.48. Suppose $\varphi : \mathbb{K}_1 \rightarrow \mathbb{K}_2$ is an isomorphism of fields and $p(x) \in \mathbb{K}_1[x]$. Suppose $\mathbb{L}_1$ is any splitting field of $p(x)$ and $\mathbb{L}_2$ is any splitting field of $\varphi(p(x))$. Then there exists a field isomorphism $\psi : \mathbb{L}_1 \rightarrow \mathbb{L}_2$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathbb{L}_1 & \xrightarrow{\psi} & \mathbb{L}_2 \\ \uparrow & & \uparrow \\ \mathbb{K}_1 & \xrightarrow{\varphi} & \mathbb{K}_2 \end{array}$$

In other words, φ can be extended to an isomorphism ψ of splitting fields.

Proof. Suppose $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{K}_1[x]$. We prove the statement by induction on n .

- If $n = 1$, then $\mathbb{L}_1 = \mathbb{K}_1$ and $\mathbb{L}_2 = \mathbb{K}_2$, thus there is nothing to show.
- Assume that the theorem holds if $\deg p < n$. If $p(x)$ splits completely over $\mathbb{K}_1$, there is nothing to prove. If we assume that $p(x)$ does not split completely over $\mathbb{K}_1$, then $\mathbb{K}_1 \subsetneq \mathbb{L}_1$, so we can choose some root $\alpha \in \mathbb{L}_1 \setminus \mathbb{K}_1$ of $p(x)$. We therefore have extensions

$$\mathbb{K}_1 \subsetneq \mathbb{K}_1(\alpha) \subseteq \mathbb{L}_1.$$

Let $m_{\alpha, \mathbb{K}_1}(x) \in \mathbb{K}_1[x]$ be the minimal polynomial of α . Then we have isomorphisms

$$\mathbb{K}_1[x]/(m_{\alpha, \mathbb{K}_1}(x)) \cong \mathbb{K}_1(\alpha),$$

and $\varphi_x(m_{\alpha, \mathbb{K}_1}(x))$ is irreducible over $\mathbb{K}_2$. Therefore $\mathbb{K}_2[x]/(\varphi_x(m_{\alpha, \mathbb{K}_1}(x)))$ is a field extension of $\mathbb{K}_2$. Note that $m_{\alpha, \mathbb{K}_1}(x) \mid p(x)$ and therefore $\varphi_x(m_{\alpha, \mathbb{K}_1}(x)) \mid \varphi_x(p(x))$. Since $\varphi_x(p(x))$ splits completely over $\mathbb{L}_2$, we have that $\varphi_x(m_{\alpha, \mathbb{K}_1}(x))$ splits completely over $\mathbb{L}_2$. Therefore there is a field homomorphism $\mathbb{K}_2[x]/(\varphi_x(m_{\alpha, \mathbb{K}_1}(x)))$ into $\mathbb{L}_2$, that sends x to a root of $\varphi_x(m_{\alpha, \mathbb{K}_1}(x))$.

In $\mathbb{K}_1(\alpha)[x]$ we have $p(x) = (x - \alpha) \cdot q(x)$ for some $q(x) \in \mathbb{F}(\alpha)[x]$. Similarly, $\varphi_x(p(x)) = (x - \alpha_1) \cdot \varphi_x(q(x))$ in $\mathbb{K}_2(\alpha_1)[x]$. Since $p(x) = (x - \alpha) \cdot q(x)$ splits completely in $\mathbb{L}_1$, $q(x)$ must split completely in $\mathbb{L}_1$. On the other hand, $q(x)$ does not split completely in any proper subextension of $\mathbb{L}_1$, since if it did, then $p(x)$ would also split completely, contradicting the minimality of $\mathbb{L}_1$ for $p(x)$. Thus $\mathbb{L}_1$ is the splitting field of $q(x) \in \mathbb{K}_1(\alpha)[x]$. Similarly, $\mathbb{L}_2$ is the splitting field of $\varphi_x(q(x)) \in \mathbb{K}_2(\alpha_1)[x]$. But $\deg q < \deg p$, so by induction hypothesis there is an isomorphism $\psi : \mathbb{L}_1 \rightarrow \mathbb{L}_2$ extending $\mathbb{K}_1(\alpha) \rightarrow \mathbb{K}_2(\alpha_1)$, which in turn extends $\varphi : \mathbb{K}_1 \rightarrow \mathbb{K}_2$. ■

Example 1.49.

1. The splitting field of $x^2 - 2 \in \mathbb{Q}[x]$ is $\mathbb{Q}(\sqrt{2}, -\sqrt{2}) = \mathbb{Q}(\sqrt{2})$. Notice that $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$.
2. Consider the polynomial $x^3 - 2 \in \mathbb{Q}[x]$. The field $\mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt[3]{2})$ contains a root of $x^3 - 2$. Therefore, in $\mathbb{Q}(\sqrt[3]{2})[x]$, it holds that

$$x^3 - 2 = (x - \sqrt[3]{2})(x^2 + \sqrt[3]{2}x + \sqrt[3]{4}).$$

The polynomial splits, but not completely. Indeed, the roots of $x^3 - 2$ are $\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2$, where $\omega = e^{\frac{2\pi i}{3}}$. The splitting field of $x^3 - 2$ therefore is $\mathbb{Q}(\sqrt[3]{2}, \omega)$. Note that, in this case, $[\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}] = 3! \neq 3$.

Lemma 1.50. Suppose $\varphi : \mathbb{K}_1 \rightarrow \mathbb{K}_2$ is a field isomorphism and $\mathbb{F}_1 \hookrightarrow \mathbb{K}_1$ is a subfield. Let $\mathbb{F}_2 := \varphi(\mathbb{F}_1)$. If $\mathbb{K}_1$ is a splitting field of some $p(x) \in \mathbb{F}_1[x] \setminus \mathbb{F}_1$, then $\mathbb{K}_2$ is a splitting field of $\varphi(p(x))$.

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Proof. Suppose $p(x) = a(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n)$ in $\mathbb{K}_1[x]$. Then

$$\varphi(p(x)) = \varphi(a)(x - \varphi(\alpha_1))(x - \varphi(\alpha_2)) \cdots (x - \varphi(\alpha_n))$$

in $\mathbb{K}_2[x]$. So $\varphi(p(x))$ splits completely over $\mathbb{F}_2$.

If $\varphi(p(x))$ splits completely over $\mathbb{L}_2$ for some extension $\mathbb{F}_2 \hookrightarrow \mathbb{L}_2 \hookrightarrow \mathbb{K}_2$. Then $p(x)$ splits completely over $\mathbb{L}_1 := \varphi^{-1}(\mathbb{L}_2)$. If $\mathbb{L}_2 \neq \mathbb{K}_2$, then $\mathbb{L}_1 \neq \mathbb{K}_1$, a contradiction. ■

We prove a generalized converse:

Theorem 1.51. *Suppose we have:*

- *A field isomorphism $\varphi : \mathbb{F}_1 \rightarrow \mathbb{F}_2$.*
- *A polynomial $p(x) \in \mathbb{F}_1[x] \setminus \mathbb{F}_1$.*
- *A splitting field $\mathbb{K}_1$ of $p(x)$ over $\mathbb{F}_1$.*
- *A field extension $\mathbb{F}_2 \hookrightarrow \mathbb{L}_2$, where $\varphi(p(x))$ splits completely.*

Then there exists a field homomorphism $\psi : \mathbb{K}_1 \rightarrow \mathbb{L}_2$, extending φ . If $\mathbb{L}_2$ is a splitting field of $\varphi(p(x))$, then $\psi : \mathbb{K}_1 \rightarrow \mathbb{L}_1$ is an isomorphism.

Proof. To prove the last statement, assuming existence of ψ , note that $\varphi(\mathbb{K}_1) \hookrightarrow \mathbb{L}_2$ is a splitting field of $\varphi(p(x))$, by the previous lemma. So, if $\mathbb{L}_2$ is a splitting field, we have $\psi(\mathbb{K}_1) = \mathbb{L}_2$, so ψ is an isomorphism.

We prove the first statement by induction on $[\mathbb{K}_1 : \mathbb{F}_1]$.

- If $[\mathbb{K}_1 : \mathbb{F}_1] = 1$, then $\mathbb{K}_1 = \mathbb{F}_1$ and there is nothing to prove.
- Suppose $[\mathbb{K}_1 : \mathbb{F}_1] = n > 1$, and an appropriate ψ exists whenever the degree of an appropriate extension is less than n . We can write $p(x) = q(x) \cdot r(x)$ in $\mathbb{F}_1[x]$, where $q(x)$ is monic and irreducible with $\deg q > 1$. Since $p(x)$ splits completely in $\mathbb{K}_1[x]$, so $q(x)$ itself splits completely in $\mathbb{K}_1[x]$. Let $\alpha \in \mathbb{K}_1 \setminus \mathbb{F}_1$ be a root of $q(x)$. Similarly, $\varphi(q(x))$ is irreducible over $\mathbb{F}_2[x]$ and splits completely over $\mathbb{L}_1$, so let β be a root of $\varphi(q(x))$ in $\mathbb{L}_1$. The polynomials $q(x)$ and $\varphi(q(x))$ are minimal polynomials of α and β , respectively. We have a field isomorphisms

$$\mathbb{F}_1[x]/(q(x)) \rightarrow \mathbb{F}_1(\alpha), \quad \text{and} \quad \mathbb{F}_1[x]/(\varphi(q(x))) \rightarrow \mathbb{F}_2(\beta),$$

which, together with φ , induce a field isomorphism $\varphi' : \mathbb{F}_1(\alpha) \rightarrow \mathbb{F}_2(\beta)$. Note that $\mathbb{K}_1$ is a splitting field for $p(x)$ over $\mathbb{F}_1(\alpha)$. Moreover,

$$[\mathbb{K}_1 : \mathbb{F}_1(\alpha)] = \frac{[\mathbb{K}_1 : \mathbb{F}_1]}{[\mathbb{F}_1(\alpha) : \mathbb{F}_1]} = \frac{[\mathbb{K}_1 : \mathbb{F}_1]}{\deg q} < [\mathbb{K}_1 : \mathbb{F}_1].$$

By the induction hypothesis, there exists a homomorphism $\psi : \mathbb{K}_1 \rightarrow \mathbb{L}_2$, extending φ' and therefore also φ . ■

Lemma 1.52. Let $p(x) \in \mathbb{F}[x]$ with $\deg p = n$, and let $\mathbb{K}$ be a splitting field of $p(x)$. Then

$$[\mathbb{K} : \mathbb{F}] \leq n!.$$

1.4. Splitting fields for a family of polynomials

Definition 1.53. Let $\mathbb{F}$ be a field, let $P = \{p_\alpha(x) : \alpha \in I\} \subset \mathbb{F}[x] \setminus \mathbb{F}$ be a family of polynomials. A splitting field for P is an extension $\mathbb{F} \hookrightarrow \mathbb{K}$, such that

1. Every polynomial $p_\alpha(x), \alpha \in P$ splits completely over $\mathbb{K}$.
2. $\mathbb{K}$ is minimal with respect to the first property.

Remark 1.54. If P only consists of finitely many, say $P = \{p_1(x), \dots, p_n(x)\}$, then the splitting field for P is the same as the splitting field for $p_1(x) \cdots p_n(x)$.

Theorem 1.55. For any family of polynomials, there exists a splitting field that is unique up to isomorphism of extensions.

Proof. Let us prove existence when P is a countably infinite family, i.e. $P = \{p_1(x), p_2(x), \dots\} \subset \mathbb{F}[x] \setminus \mathbb{F}$. Construct a sequence of extensions

$$\mathbb{F} = \mathbb{K}_0 \hookrightarrow \mathbb{K}_1 \hookrightarrow \mathbb{K}_2 \hookrightarrow \dots \hookrightarrow \mathbb{K}_n \hookrightarrow \dots,$$

where each $\mathbb{K}_n$ is the splitting for $p_n(x)$ over $\mathbb{K}_{n-1}$. Let

$$\mathbb{K} := \bigcup_{n=0}^{\infty} \mathbb{K}_n.$$

It is easily checked that $\mathbb{K}$ is indeed a field, and that each $p_n(x)$ splits completely over $\mathbb{K}$. We now claim that $\mathbb{K}$ is already the splitting field, which means it remains to show that it is minimal.

Suppose we have a proper sub-extension $\mathbb{F} \hookrightarrow \mathbb{L} \hookrightarrow \mathbb{K}$. Then there exists an $n \in \mathbb{N}$ such that $\mathbb{K}_{n-1} \subseteq \mathbb{L}$, but $\mathbb{K}_n \not\subseteq \mathbb{L}$. Then $\mathbb{L}$ is an extension of $\mathbb{K}_{n-1}$, but $\mathbb{L}$ does not contain a splitting field of $p_n(x)$ over $\mathbb{K}_{n-1}$ (since that splitting field is $\mathbb{K}_n$).

If P is uncountable, one can use Zorn's Lemma.

For uniqueness, suppose that $\mathbb{L}_1$ and $\mathbb{L}_2$ are splitting fields for P . Consider the set

$$X := \{(\mathbb{K}_1, \mathbb{K}_2, \varphi) : \mathbb{F} \hookrightarrow \mathbb{K}_1 \hookrightarrow \mathbb{L}_1, \mathbb{F} \hookrightarrow \mathbb{K}_2 \hookrightarrow \mathbb{L}_2, \varphi : \mathbb{K}_1 \rightarrow \mathbb{K}_2 \text{ is isomorphism}\}.$$

The set X can be partially ordered by considering extensions, and satisfies Zorn's Lemma. Therefore, there exists a maximal element $(\mathbb{K}_1^{\max}, \mathbb{K}_2^{\max}, \varphi^{\max})$. If $\mathbb{K}_1^{\max} \subsetneq \mathbb{L}_1$, then there exists a $p(x) \in P$ that does not split completely over $\mathbb{K}_1^{\max}$. Therefore we have splitting fields $\mathbb{K}_1^{\max} \subsetneq \mathbb{K}_1 \subseteq \mathbb{L}_1, \mathbb{K}_2^{\max} \subsetneq \mathbb{K}_2 \subseteq \mathbb{L}_2$ (for $p(x)$ and $\varphi^{\max}(p(x))$, respectively). So $\varphi^{\max}$ extends to an isomorphism $\mathbb{K}_1 \rightarrow \mathbb{K}_2$. ■

Definition 1.56. Let $\mathbb{F}$ be a field. A splitting field of the family $\mathbb{F}[x] \setminus \mathbb{F}$ is called an *algebraic closure* of $\mathbb{F}$, and is often denoted $\overline{\mathbb{F}}$.

In particular, we now know that every field $\mathbb{F}$ has an algebraic closure, which is unique up to isomorphism.

Definition 1.57. A field $\mathbb{L}$ is *algebraically closed*, if every non-constant polynomial in $\mathbb{L}[x]$ splits completely over $\mathbb{L}$. Equivalently, every such polynomial has a root in $\mathbb{L}$.

Theorem 1.58. *The field $\mathbb{C}$ is algebraically closed.*

Theorem 1.59. *Suppose we have a field extension $\mathbb{F} \hookrightarrow \mathbb{K}$, then the following are equivalent:*

1. $\mathbb{K}$ is an algebraic closure of $\mathbb{F}$.
2. Every non-constant polynomial over $\mathbb{F}$ splits completely over $\mathbb{K}$ and $\mathbb{K}$ an algebraic extension of $\mathbb{F}$.
3. $\mathbb{K}$ is algebraically closed and $\mathbb{K}$ is an algebraic extension of $\mathbb{F}$.

1.5. Separability

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Let $\mathbb{F}$ be a field and $p(x) \in \mathbb{F}[x] \setminus \mathbb{F}$. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be an extension where $p(x)$ splits completely. Thus

$$p(x) = a(x - \alpha_1) \cdots (x - \alpha_n), \quad a \in \mathbb{F}^*, \quad \alpha_1, \dots, \alpha_n \in \mathbb{K},$$

where the roots are not necessarily distinct.

Definition 1.60. A polynomial $p(x) \in \mathbb{F}[x]$ is called *separable* if in some (and therefore any) extension of $\mathbb{F}$ where $p(x)$ splits completely, the roots of $p(x)$ are pairwise distinct.

Example 1.61.

1. The polynomial $x^2 - 2x + 1 \in \mathbb{Q}[x]$ is not separable.

2. The polynomial $x^2 - 2 \in \mathbb{Q}[x]$ factors as $(x - \sqrt{2})(x + \sqrt{2})$, and is therefore separable.

Does there exist an irreducible polynomial over $p(x) \in \mathbb{Q}[x]$ that is not separable? We will see later that this is not the case.

Definition 1.62. Let $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{F}[x]$. We define the *formal derivative* of $p(x)$ by

$$p'(x) := D(p(x)) := a_1 + 2a_2x + \dots + ka_kx^{k-1} + \dots + na_nx^{n-1}.$$

Lemma 1.63. *It holds that:*

- $D(p(x) + q(x)) = D(p(x)) + D(q(x))$
- $D(p(x)q(x)) = D(p(x))q(x) + D(q(x))p(x)$
- $D(p(q(x))) = p'(q(x))q'(x)$
- $\deg p'(x) \leq \deg p(x) - 1$

Lemma 1.64. *Let $p(x) \in \mathbb{F}[x]$ and $\mathbb{F} \hookrightarrow \mathbb{K}$. An element $\alpha \in \mathbb{K}$ is a multiple root of $p(x)$ in $\mathbb{K}[x]$ if and only if $(x - \alpha) \mid p(x)$ and $(x - \alpha) \mid D(p(x))$ in $\mathbb{K}[x]$.*

Proof. Suppose α is a multiple root. Then $p(x) = (x - \alpha)^2q(x)$ for some polynomial $q(x) \in \mathbb{K}[x]$. But then

$$D(p(x)) = 2(x - \alpha)q(x) + (x - \alpha)^2D(q(x)) = (x - \alpha)(2q(x) + (x - \alpha)D(q(x))),$$

hence $(x - \alpha) \mid D(p(x))$.

Conversely, suppose $(x - \alpha)$ divides both $p(x)$ and $D(p(x))$. Then we can write $p(x) = (x - \alpha)q(x)$, hence

$$D(p(x)) = q(x) + (x - \alpha)D(q(x)).$$

Since $(x - \alpha) \mid D(p(x))$ it also divides $q(x)$, therefore

$$p(x) = (x - \alpha)^2r(x)$$

and α is a multiple root. ■

Lemma 1.65. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $p(x), q(x) \in \mathbb{F}[x]$. Then*

$$\gcd_{\mathbb{F}[x]}(p(x), q(x)) = \gcd_{\mathbb{K}[x]}(p(x), q(x)).$$

Proof. Recall that a polynomial $d(x) \in \mathbb{F}[x]$ is a gcd of $p(x)$ and $q(x)$ if and only if (a) $d(x) \mid p(x)$ and $d(x) \mid q(x)$ in $\mathbb{F}[x]$ and (b) there exist $s(x), t(x) \in \mathbb{F}[x]$ such that $d(x) = s(x)p(x) + t(x)q(x)$. So let $d(x) = \gcd_{\mathbb{F}[x]}(p(x), q(x))$, then it is immediate that $d(x)$ satisfies (a) and (b) over $\mathbb{K}$ as well. ■

Proposition 1.66. *A polynomial $p(x) \in \mathbb{F}[x]$ is separable if and only if it is relatively prime (in $\mathbb{F}[x]$) to its derivative.*

Proof. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be an extension where $p(x)$ splits completely, so

$$p(x) = a(x - \alpha_1) \cdots (x - \alpha_n), \quad a \in \mathbb{F}^*, \quad \alpha_1, \dots, \alpha_n \in \mathbb{K}.$$

By the previous lemma, it is enough to show that $\alpha_1, \dots, \alpha_n$ are pairwise distinct if and only if $\gcd_{\mathbb{K}[x]}(p(x), D(p(x))) = 1$. If they are not distinct, then $p(x)$ has a multiple root α , and we have shown previously that $(x - \alpha) \mid \gcd_{\mathbb{K}[x]}(p(x), D(p(x)))$. If they are distinct, then an easy calculation shows that $\alpha_1, \dots, \alpha_n$ are not roots of $D(p(x))$. So $p(x)$ and $D(p(x))$ have no common roots. But since $p(x)$ splits completely in $\mathbb{K}[x]$, it follows that $\gcd_{\mathbb{K}[x]}(p(x), D(p(x))) = 1$. ■

Corollary 1.67. *Suppose $p(x) \in \mathbb{F}[x]$ is irreducible. Then $p(x)$ is separable if and only if $D(p(x)) \neq 0$.*

Proof. If $p(x)$ is irreducible, then $\gcd(p(x), D(p(x)))$ is either 1 or $p(x)$. But $\deg D(p(x)) < \deg p(x)$, therefore $p(x)$ cannot be a divisor of $D(p(x))$, unless $D(p(x)) = 0$, in which case $\gcd(p(x), D(p(x))) = p(x)$. ■

Corollary 1.68. *If $\text{char } \mathbb{F} = 0$, then any irreducible polynomial over $\mathbb{F}$ is separable.*

Definition 1.69. A field $\mathbb{F}$ is called *perfect* if every irreducible polynomial over $\mathbb{F}$ is also separable.

Definition 1.70. An algebraic extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is called *separable* if for every $\alpha \in \mathbb{K}$, the minimal polynomial $m_{\alpha, \mathbb{K}}(x)$ is separable.

Example 1.71. Suppose $\mathbb{E}$ is a field of characteristic $p \in \mathbb{P}$, and $\alpha \in \mathbb{E}$ such that

$$\eta^p \neq \alpha, \quad \forall \eta \in \mathbb{E}.$$

Consider the polynomial $q(x) := x^p - \alpha$. Then $D(q(x)) = px^{p-1} = 0$, so $q(x)$ is not separable. We can also verify this manually: let $\mathbb{E} \hookrightarrow \mathbb{L}$ be an extension where $q(x)$ splits completely. So there exists an element $\alpha^{\frac{1}{p}} \in \mathbb{L}$ such that $(\alpha^{\frac{1}{p}})^p = \alpha \in \mathbb{E}$. In $\mathbb{L}[x]$, it holds that

$$x^p - \alpha = x^p - (\alpha^{\frac{1}{p}})^p = (x - \alpha^{\frac{1}{p}})^p.$$

We now want to show that $x^p - \alpha$ is irreducible in $\mathbb{E}[x]$. Suppose $x^p - \alpha = f(x)g(x)$ where $f(x) \in \mathbb{E}[x]$ is irreducible with $1 \leq \deg f(x) < p$. In $\mathbb{L}[x]$, the decomposition has the form

$$(x - \alpha^{\frac{1}{p}})^p = (x - \alpha^{\frac{1}{p}})^a (x - \alpha^{\frac{1}{p}})^b,$$

where $1 \leq a < p$, therefore $(x - \alpha^{\frac{1}{p}})^a \in \mathbb{E}[x]$. This shows that $(\alpha^{\frac{1}{p}})^a \in \mathbb{E}$. Since $1 \leq a < p$ we can find $s, t \in \mathbb{Z}$ such that $sa + tp = 1$. Therefore

$$\alpha^{\frac{1}{p}} = (\alpha^{\frac{1}{p}})^{sa+tp} \in \mathbb{E},$$

a contradiction to α not being a p -th power.

Example 1.72. Let $\mathbb{F}_p(t)$ be the quotient field of $\mathbb{F}_p[t]$. The elements of this field are expressions of the form

$$\frac{f(t)}{g(t)}, \quad f(t), g(t) \in \mathbb{F}_p[t], \quad g(t) \neq 0.$$

In this field, not every element is a p -th power. Indeed,

$$\left(\frac{a_0 + a_1 t + \dots + a_n t^n}{b_0 + b_1 t + \dots + b_m t^m} \right)^p = \frac{a_0^p + a_1^p t^p + \dots + a_n^p t^{np}}{b_0^p + b_1^p t^p + \dots + b_m^p t^{mp}}.$$

So, by the previous example, $x^p - t$ is an irreducible non-separable polynomial in $\mathbb{F}_p(t)[x]$.

Definition 1.73. Suppose $\text{char } \mathbb{F} = p$. Define the *Frobenius map* by

$$\sigma : \mathbb{F} \rightarrow \mathbb{F}, \alpha \mapsto \alpha^p.$$

Since the field has characteristic p , this map is a field monomorphism.

Theorem 1.74. A field $\mathbb{F}$ of characteristic $p \in \mathbb{P}$ is perfect if and only if the Frobenius map is an automorphism.

Proof. We saw in the previous examples that, if σ is not surjective, then $\mathbb{F}$ is not perfect.

Conversely, suppose that σ is surjective, in particular an automorphism. Suppose towards a contradiction that $q(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{F}[x]$ is irreducible and non-separable. By a previous lemma, this implies that $D(q(x)) = 0$, therefore $\ell a_\ell = 0$ in $\mathbb{F}$ for all $\ell = 1, \dots, n$. It follows that $a_k = 0$ whenever $p \nmid k$.

So $q(x) = a_0 + a_px^p + a_{2p}x^{2p} + \dots + a_{kp}x^{kp}$. Since σ is surjective, we can write $a_{\ell p} = b_\ell^p$ for some $b_\ell \in \mathbb{F}$, for $\ell = 0, \dots, k$. Therefore

$$q(x) = b_0^p + (b_1x)^p + (b_2x^2)^p + \dots (b_kx^k)^p = (b_0 + b_1x + \dots + b_kx^k)^p$$

and $q(x)$ is not irreducible, a contradiction. ■

Corollary 1.75. *A finite field is perfect.*

Theorem 1.76. *For every $p \in \mathbb{P}$ and $k \in \mathbb{N}$ there exists a field, which we will denote $\mathbb{F}_{p^k}$ with p^k elements. Any two fields with p^k elements are isomorphic.*

Proof. Consider the polynomial $q(x) := x^{p^k} - x \in \mathbb{F}_p[x]$ and satisfies $D(q(x)) = -1$, and is therefore separable. Let $\mathbb{F}_p \hookrightarrow \mathbb{K}$ be a splitting field of this polynomial. In $\mathbb{K}$, there are p^k distinct elements $\alpha_1, \dots, \alpha_{p^k}$ that are roots of $q(x)$. We claim that these elements already constitute a subfield of $\mathbb{K}$.

- Clearly 0 and 1 are roots.
- If α, β are roots, then $(\alpha \pm \beta)^{p^k} = \alpha^{p^k} \pm \beta^{p^k} = \alpha \pm \beta$, as well as $(\alpha\beta)^{p^k} = \alpha\beta$ and $\left(\frac{\alpha}{\beta}\right)^{p^k} = \frac{\alpha}{\beta}$.

In this subfield $q(x)$ splits completely, and by minimality of $\mathbb{K}$ it follows that

$$\mathbb{K} = \{\alpha_1, \dots, \alpha_{p^k}\}.$$

Therefore we can define $\mathbb{F}_{p^k}$ as the splitting field of $q(x)$ over $\mathbb{F}_p$.

On the other hand, if $\mathbb{E}$ is another field with p^k elements, then every $\alpha \in \mathbb{E}$ satisfies $\alpha^{p^k} = \alpha$, in other words, α is a root of $q(x)$. So $q(x)$ splits completely over $\mathbb{E}$, and since it cannot split in any smaller field (there are not enough elements) it follows that $\mathbb{E}$ is a splitting field of $q(x)$. ■

1.6. Field homomorphisms (embeddings) and automorphisms

Let $\mathbb{K}$ be a field and $\varphi : \mathbb{K} \rightarrow \mathbb{K}$ a field homomorphism^[3]. Define the *field field* by

$$\mathbb{K}^\varphi := \{x \in \mathbb{K} : \varphi(x) = x\}.$$

Lemma 1.77. $\mathbb{K}^\varphi$ is a subfield of $\mathbb{K}$.

Proof. Follows immediately by computation. ■

Let $\text{Aut}(\mathbb{K})$ be the group of automorphisms of $\mathbb{K}$ and suppose $H \subseteq \text{Aut}(\mathbb{K})$. Define

$$\mathbb{K}^H := \{x \in \mathbb{K} : \varphi(x) = x \text{ for all } \varphi \in H\} = \bigcap_{\varphi \in H} \mathbb{K}^\varphi.$$

By the above lemma, it follows that $\mathbb{K}^H$ is a subfield of $\mathbb{K}$.

Example 1.78. Complex conjugation $c : \mathbb{C} \rightarrow \mathbb{C}, z \mapsto \bar{z}$ is a field automorphism, with $\mathbb{C}^c = \mathbb{R}$. Furthermore $\{\text{id}, c\} \cong \mathbb{Z}_2$ is a group acting on $\mathbb{C}$ by automorphisms, and it holds that $\mathbb{C}^{\{\text{id}, c\}} = \mathbb{R}$.

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension. Define

$$\text{Aut}_{\mathbb{F}}(\mathbb{K}) := \text{Gal}_{\mathbb{F}}(\mathbb{K}) := G_{\mathbb{F}}(\mathbb{K}) := \{\varphi \in \text{Aut}(\mathbb{K}) : \varphi(x) = x \text{ for all } x \in \mathbb{F}\}.$$

Lemma 1.79. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$, $\varphi \in \text{Gal}_{\mathbb{F}}(\mathbb{K})$, $p(x) \in \mathbb{F}[x]$ and $\alpha \in \mathbb{K}$ is a root of $p(x)$. Then $\varphi(\alpha)$ is also a root of $p(x)$.

Proof. Suppose $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{F}[x]$. Then

$$\begin{aligned} p(\varphi(\alpha)) &= a_0 + a_1\varphi(\alpha) + \dots + a_n\varphi(\alpha)^n \\ &= \varphi(a_0) + \varphi(a_1)\varphi(\alpha) + \dots + \varphi(a_n)\varphi(\alpha^n) \\ &= \varphi(p(\alpha)) = 0. \end{aligned} \quad \blacksquare$$

Corollary 1.80. The group $\text{Gal}_{\mathbb{F}}(\mathbb{K})$ acts by permutation on the set of roots of any polynomial over $\mathbb{F}$.

Corollary 1.81. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is an algebraic extension. Then $\text{Gal}_{\mathbb{F}}(\mathbb{K}) = \text{End}_{\mathbb{F}}(\mathbb{K})$.

Proof. Let $\varphi : \mathbb{K} \rightarrow \mathbb{K}$ be a homomorphism fixing $\mathbb{F}$. We know that φ is injective, we only need to show surjectivity. Let $\alpha \in \mathbb{K}$, then φ permutes the set of roots of $m_{\alpha, \mathbb{F}}(x)$. Therefore there exists a root $\beta \in \mathbb{K}$ of $m_{\alpha, \mathbb{K}}(x)$ such that $\varphi(\beta) = \alpha$. ■

^[3]Recall that φ is automatically a monomorphism, i.e. an embedding.

Remark 1.82. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension. We obtain a function Φ from the set of subgroups of $\text{Gal}_{\mathbb{F}}(\mathbb{K})$ to the set of intermediate fields between $\mathbb{F}$ and $\mathbb{K}$, via $\Phi(H) := \mathbb{K}^H$ for a subgroup H of $\text{Gal}_{\mathbb{F}}(\mathbb{K})$. Conversely, we also obtain a function Γ in the other direction, by $\Gamma(\mathbb{E}) = \text{Gal}_{\mathbb{E}}(\mathbb{K}) = \{\varphi \in \text{Gal}_{\mathbb{F}}(\mathbb{K}) : \varphi(x) = x \text{ for all } x \in \mathbb{E}\}$ for an intermediate field $\mathbb{F} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{K}$.

However, in general Φ and Γ are not inverses.

Example 1.83. We consider the extension $\mathbb{Q} \hookrightarrow \mathbb{Q}(\sqrt[3]{2}) \hookrightarrow \mathbb{R}$. What is $\text{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$? More generally: Suppose $\mathbb{Q} \hookrightarrow \mathbb{L}$ is any extension, what is $\text{Hom}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}), \mathbb{L})$?

Answer: We have

$$\begin{aligned} \text{Hom}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}), \mathbb{L}) &= \text{Hom}_{\mathbb{Q}}(\mathbb{Q}[x]/(x^3 - 2), \mathbb{L}) \\ &= \{\beta \in \mathbb{L} : \beta^3 = 2\}. \end{aligned}$$

Therefore $\text{Hom}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}), \mathbb{R}) = \{1\}$ and $\text{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2})) = \{1\}$. However, $\Gamma(\mathbb{Q}) = \{1\}$ and $\Phi(\{1\}) = (\mathbb{Q}(\sqrt[3]{2}))^{\{1\}} = \mathbb{Q}(\sqrt[3]{2})$.

Remark 1.84. We have the following “formal” properties of Φ and Γ :

1. i. If $H_1 \subseteq H_2 \subseteq \text{Gal}_{\mathbb{F}}(\mathbb{K})$, then $\Phi(H_1) \supseteq \Phi(H_2)$.
 ii. If $\mathbb{F} \hookrightarrow \mathbb{E}_1 \hookrightarrow \mathbb{E}_2 \hookrightarrow \mathbb{K}$, then $\Gamma(\mathbb{E}_1) \supseteq \Gamma(\mathbb{E}_2)$.
2. i. For every $H \subseteq \text{Gal}_{\mathbb{F}}(\mathbb{K})$, it holds that $H \subseteq \Gamma(\Phi(H))$.
 ii. For every $\mathbb{F} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{K}$, it holds that $\mathbb{E} \hookrightarrow \Phi(\Gamma(\mathbb{E}))$.
3. i. For every element $\mathbb{E} \in \text{im } \Phi$, it holds that $\Phi(\Gamma(\mathbb{E})) = \mathbb{E}$.
 ii. For every element $H \in \text{im } \Gamma$, it holds that $\Gamma(\Phi(H)) = H$.

To see the last property, suppose that $\mathbb{E} = \Phi(H)$ for some subgroup H . Then

$$\Phi(H) \subseteq \Phi(\Gamma(\Phi(H))),$$

but also $H \subseteq \Gamma(\Phi(H))$, therefore $\Phi(H) \supseteq \Phi(\Gamma(\Phi(H)))$. Therefore

$$\mathbb{E} = \Phi(H) = \Phi(\Gamma(\Phi(H))) = \Phi(\Gamma(\mathbb{E})).$$

So, while Γ and Φ are not inverses in general, they are bijections between their images.

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension. In general, by the above,

$$\mathbb{F} \subseteq \mathbb{K}^{\text{Gal}_{\mathbb{F}}(\mathbb{K})} \subseteq \mathbb{K}.$$

Theorem 1.85. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is an algebraic extension. Then the following are equivalent:

1. $\mathbb{F} = \mathbb{K}^{\text{Gal}_{\mathbb{F}}(\mathbb{K})}$
2. The extension is both normal and separable.

Recall that an extension being normal means that it is the splitting field of a family of polynomials over $\mathbb{F}$, or equivalently, the minimal polynomial of every $\alpha \in \mathbb{K}$ splits completely over $\mathbb{K}$.

The extension being separable means that the minimal polynomial of every $\alpha \in \mathbb{K}$ splits into distinct linear factors over the splitting field.

Proof. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is normal and separable. Let $\alpha \in \mathbb{K} \setminus \mathbb{F}$. We need to show that there is a homomorphism $\varphi : \mathbb{K} \rightarrow \mathbb{K}$ fixing $\mathbb{F}$ pointwise but not α . Consider the minimal polynomial of $m_{\alpha, \mathbb{F}}(x) \in \mathbb{F}[x]$, which has degree ≥ 2 . By assumption, $m_{\alpha, \mathbb{F}}(x)$ has another root $\beta \in \mathbb{K}$ with $\beta \neq \alpha$, then $m_{\alpha, \mathbb{F}}(x) = m_{\beta, \mathbb{F}}(x)$. It follows that the fields $\mathbb{F}(\alpha)$ and $\mathbb{F}(\beta)$ are isomorphic extensions of $\mathbb{F}$. Moreover, there is a (unique) isomorphism $\varphi_0 : \mathbb{F}(\alpha) \rightarrow \mathbb{F}(\beta)$ which fixes $\mathbb{F}$ and satisfies $\varphi_0(\alpha) = \beta$. We obtain an embedding $\mathbb{F}(\alpha) \hookrightarrow \mathbb{F}(\beta) \subseteq \mathbb{K}$. Now, $\mathbb{K}$ is a splitting field over $\mathbb{F}(\alpha)$, therefore any embedding $\varphi_0 : \mathbb{F}(\alpha) \rightarrow \mathbb{K}$ fixing $\mathbb{F}$ extends to a homomorphism $\varphi : \mathbb{K} \rightarrow \mathbb{K}$ satisfying $\varphi(\alpha) = \beta \neq \alpha$.

Now suppose that $\mathbb{F} = \mathbb{K}^{\text{Gal}_{\mathbb{F}}(\mathbb{K})}$. Let $\alpha \in \mathbb{K}$ and $m_{\alpha, \mathbb{F}}(x)$ be the minimal polynomial of α . Let $\alpha_1, \dots, \alpha_n$ denote the distinct roots of $m_{\alpha, \mathbb{F}}(x)$ and suppose $\alpha = \alpha_1$. The group $\text{Gal}_{\mathbb{F}}(\mathbb{K})$ acts on $\{\alpha_1, \dots, \alpha_n\}$ by permutation. Consider the polynomial

$$p(x) = (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n) \in \mathbb{K}[x].$$

For any $\varphi \in \text{Gal}_{\mathbb{F}}(\mathbb{K})$, it holds that $\varphi(p(x)) = p(x)$. Therefore

$$p(x) \in \mathbb{K}^{\text{Gal}_{\mathbb{F}}(\mathbb{K})}[x] = \mathbb{F}[x].$$

It follows that $m_{\alpha, \mathbb{F}}(x) \mid p(x)$. In fact $p(x) = m_{\alpha, \mathbb{F}}(x)$, because each α_k is a root of $m_{\alpha, \mathbb{F}}(x)$. In particular, $m_{\alpha, \mathbb{F}}(x)$ splits as a product of distinct linear factors. ■

Definition 1.86. An algebraic extension is *Galois* if it satisfies any of the (equivalent) conditions of the theorem above.

Corollary 1.87. *If $\mathbb{F} \hookrightarrow \mathbb{K}$ is a Galois extension (i.e. algebraic, normal and separable), then Φ is injective and Γ is surjective.*

Proof. If $\mathbb{F} \hookrightarrow \mathbb{K}$ is normal and separable, then for every intermediate extension $\mathbb{E}$ it holds that also $\mathbb{K}$ is normal and separable over $\mathbb{E}$. So $\mathbb{E} = \mathbb{K}^{\text{Gal}_{\mathbb{E}}(\mathbb{K})} = \Phi(\Gamma(\mathbb{E}))$. ■

Theorem 1.88. *Let $\mathbb{K}$ be a field and let $G \subseteq \text{Aut}(\mathbb{K})$ be a finite subgroup. Then*

$$[\mathbb{K} : \mathbb{K}^G] = |G|.$$

Corollary 1.89. *For any field extension $\mathbb{F} \hookrightarrow \mathbb{K}$, the finite subgroups of $\text{Gal}_{\mathbb{F}}(\mathbb{K})$ are in the image of Γ .*

Proof. Let G be a such a finite subgroup. We know that $G \subseteq \Gamma(\Phi(G)) = \text{Gal}_{\mathbb{K}^G}(\mathbb{K})$. On the other hand

$$|\Gamma(\Phi(G))| = [\mathbb{K} : \mathbb{K}^{\Gamma(\Phi(G))}] = [\mathbb{K} : \Phi(\Gamma(\Phi(G)))] = [\mathbb{K} : \Phi(G)] = [\mathbb{K} : \mathbb{K}^G] = |G|,$$

and therefore $G = \Gamma(\Phi(G))$. ■

Ultimately, if $\mathbb{K}/\mathbb{F}$ is a finite Galois extension, then Φ, Γ are inverse bijections.